

A5P2

October 21, 2025

```
[6]: !pip install torch torchvision torchaudio --extra-index-url https://download.pytorch.org/whl/cpu
```

```
Defaulting to user installation because normal site-packages is not writeable
Looking in indexes: https://pypi.org/simple,
https://download.pytorch.org/whl/cpu
Looking in links: /usr/share/pip-wheels
Collecting torch
  Obtaining dependency information for torch from
https://download.pytorch.org/whl/cpu/torch-2.8.0%2Bcpu-
cp311-cp311-manylinux_2_28_x86_64.whl.metadata
  Downloading https://download.pytorch.org/whl/cpu/torch-2.8.0%2Bcpu-
cp311-cp311-manylinux_2_28_x86_64.whl.metadata (29 kB)
Collecting torchvision
  Obtaining dependency information for torchvision from
https://download.pytorch.org/whl/cpu/torchvision-0.23.0%2Bcpu-
cp311-cp311-manylinux_2_28_x86_64.whl.metadata
  Downloading https://download.pytorch.org/whl/cpu/torchvision-0.23.0%2Bcpu-
cp311-cp311-manylinux_2_28_x86_64.whl.metadata (6.1 kB)
Collecting torchaudio
  Obtaining dependency information for torchaudio from
https://download.pytorch.org/whl/cpu/torchaudio-2.8.0%2Bcpu-
cp311-cp311-manylinux_2_28_x86_64.whl.metadata
  Downloading https://download.pytorch.org/whl/cpu/torchaudio-2.8.0%2Bcpu-
cp311-cp311-manylinux_2_28_x86_64.whl.metadata (7.2 kB)
Requirement already satisfied: filelock in /opt/conda/envs/anaconda-
panel-2023.05-py310/lib/python3.11/site-packages (from torch) (3.9.0)
Requirement already satisfied: typing-extensions>=4.10.0 in
./local/lib/python3.11/site-packages (from torch) (4.15.0)
Requirement already satisfied: sympy>=1.13.3 in ./local/lib/python3.11/site-
packages (from torch) (1.14.0)
Requirement already satisfied: networkx in /opt/conda/envs/anaconda-
panel-2023.05-py310/lib/python3.11/site-packages (from torch) (3.1)
Requirement already satisfied: jinja2 in /opt/conda/envs/anaconda-
panel-2023.05-py310/lib/python3.11/site-packages (from torch) (3.1.2)
Requirement already satisfied: fsspec in /opt/conda/envs/anaconda-
panel-2023.05-py310/lib/python3.11/site-packages (from torch) (2023.4.0)
Requirement already satisfied: numpy in /opt/conda/envs/anaconda-
```

panel-2023.05-py310/lib/python3.11/site-packages (from torchvision) (1.24.3)
Requirement already satisfied: pillow!=8.3.*,>=5.3.0 in
/opt/conda/envs/anaconda-panel-2023.05-py310/lib/python3.11/site-packages (from
torchvision) (9.4.0)
Requirement already satisfied: mpmath<1.4,>=1.1.0 in /opt/conda/envs/anaconda-
panel-2023.05-py310/lib/python3.11/site-packages (from sympy>=1.13.3->torch)
(1.3.0)
Requirement already satisfied: MarkupSafe>=2.0 in /opt/conda/envs/anaconda-
panel-2023.05-py310/lib/python3.11/site-packages (from jinja2->torch) (2.1.1)
Downloading https://download.pytorch.org/whl/cpu/torch-2.8.0%2Bcpu-
cp311-cp311-manylinux_2_28_x86_64.whl (184.1 MB)
184.1/184.1
MB 6.2 MB/s eta 0:00:00 0:00:01[36m0:00:01
Downloading https://download.pytorch.org/whl/cpu/torchvision-0.23.0%2Bcpu-
cp311-cp311-manylinux_2_28_x86_64.whl (1.9 MB)
1.9/1.9
MB 37.4 MB/s eta 0:00:00 0:00:01
Downloading https://download.pytorch.org/whl/cpu/torchaudio-2.8.0%2Bcpu-
cp311-cp311-manylinux_2_28_x86_64.whl (1.8 MB)
1.8/1.8
MB 36.2 MB/s eta 0:00:00 0:00:01
Installing collected packages: torch, torchvision, torchaudio
WARNING: The scripts torchfrtrace and torchrun are installed in
'/home/c821c172-eb2e-4c76-bda4-16056937fc04/.local/bin' which is not on PATH.
Consider adding this directory to PATH or, if you prefer to suppress this
warning, use --no-warn-script-location.
Successfully installed torch-2.8.0+cpu torchaudio-2.8.0+cpu
torchvision-0.23.0+cpu

```
[1]: import torch
import torchvision
import torchvision.transforms as transforms
import matplotlib.pyplot as plt
import numpy as np
```

```
[2]: transform = transforms.Compose(
    [transforms.ToTensor(),
     transforms.Normalize((0.5,), (0.5,))])

batch_size = 4

trainset = torchvision.datasets.MNIST(root='./data', train=True,
                                     download=True, transform=transform)
```

```

trainloader = torch.utils.data.DataLoader(trainset, batch_size=batch_size,
                                           shuffle=True, num_workers=2)

testset = torchvision.datasets.MNIST(root='./data', train=False,
                                      download=True, transform=transform)
testloader = torch.utils.data.DataLoader(testset, batch_size=batch_size,
                                          shuffle=False, num_workers=2)

classes = ('0', '1', '2', '3',
           '4', '5', '6', '7', '8', '9')

```

```

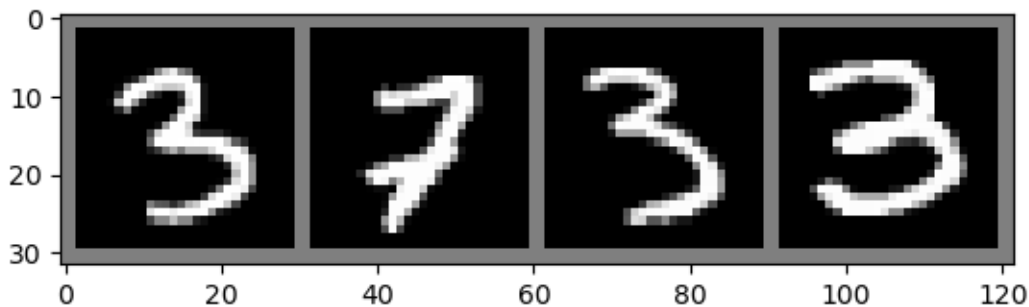
[3]: def imshow(img):
      img = img / 2 + 0.5
      npimg = img.numpy()
      if npimg.shape[0] == 1:
          plt.imshow(npimg[0], cmap='gray')
      else:
          plt.imshow(np.transpose(npimg, (1, 2, 0)))
      plt.show()

      dataiter = iter(trainloader)
      images, labels = next(dataiter)

      imshow(torchvision.utils.make_grid(images))

      print(' '.join(f'{classes[labels[j]]:5s}' for j in range(batch_size)))

```



3 7 3 3

Traceback (most recent call last):

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File "/opt/conda/envs/anaconda-panel-2023.05-py310/lib/python3.11/multiprocessing/util.py", line 300, in _run_finalizers
 finalizer()

```

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OSError: [Errno 16] Device or resource busy: '.nfs00000000053d2d6800003951'
OSError: [Errno 16] Device or resource busy: '.nfs00000000053d2d6700003950'

```

```

[4]: import torch.nn as nn
import torch.nn.functional as F

```

```

class Net(nn.Module):
    def __init__(self):
        super().__init__()
        self.conv1 = nn.Conv2d(1, 6, 5)
        self.pool = nn.MaxPool2d(2, 2)
        self.conv2 = nn.Conv2d(6, 16, 5)
        self.fc1 = nn.Linear(16 * 4 * 4, 120)
        self.fc2 = nn.Linear(120, 84)
        self.fc3 = nn.Linear(84, 10)

    def forward(self, x):
        x = self.pool(F.relu(self.conv1(x)))
        x = self.pool(F.relu(self.conv2(x)))
        x = torch.flatten(x, 1)
        x = F.relu(self.fc1(x))
        x = F.relu(self.fc2(x))
        x = self.fc3(x)
        return x

net = Net()

```

```

[5]: import torch.optim as optim

criterion = nn.CrossEntropyLoss()
optimizer = optim.SGD(net.parameters(), lr=0.001, momentum=0.9)

```

```

[6]: for epoch in range(2):

    running_loss = 0.0
    for i, data in enumerate(trainloader, 0):
        inputs, labels = data

        optimizer.zero_grad()

        outputs = net(inputs)
        loss = criterion(outputs, labels)
        loss.backward()
        optimizer.step()

        running_loss += loss.item()
        if i % 2000 == 1999:
            print(f'[{epoch + 1}, {i + 1:5d}] loss: {running_loss / 2000:.3f}')
            running_loss = 0.0

print('Finished Training')

```

```
[1, 2000] loss: 1.253
[1, 4000] loss: 0.216
[1, 6000] loss: 0.137
[1, 8000] loss: 0.113
[1, 10000] loss: 0.101
[1, 12000] loss: 0.087
[1, 14000] loss: 0.090
```

```
Traceback (most recent call last):
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  File "/opt/conda/envs/anaconda-
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OSError: [Errno 16] Device or resource busy: '.nfs00000000053d2d690000392f'
OSError: [Errno 16] Device or resource busy: '.nfs00000000053d2d6a00003930'

[2, 2000] loss: 0.065
[2, 4000] loss: 0.063
[2, 6000] loss: 0.062
[2, 8000] loss: 0.063
[2, 10000] loss: 0.065
[2, 12000] loss: 0.064
[2, 14000] loss: 0.059

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OSError: [Errno 16] Device or resource busy: '.nfs00000000053d2d6a0000394e'
OSError: [Errno 16] Device or resource busy: '.nfs00000000053d2d690000394d'

```

Finished Training

```

[7]: PATH = './mnist_net.pth'
    torch.save(net.state_dict(), PATH)

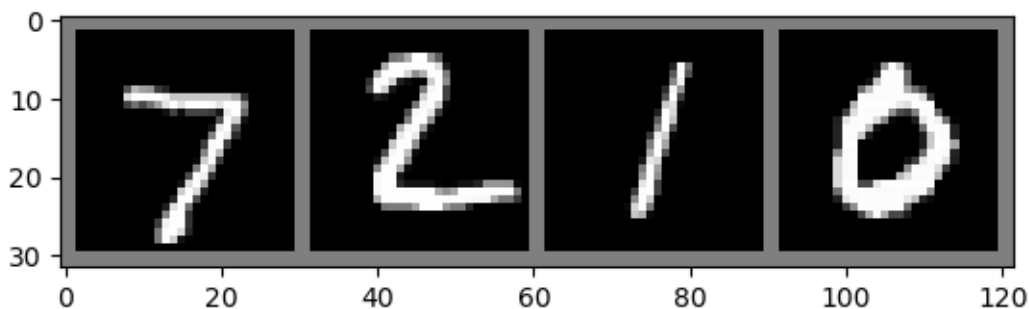
```

```

[8]: dataiter = iter(testloader)
    images, labels = next(dataiter)

    imshow(torchvision.utils.make_grid(images))
    print('GroundTruth: ', ' '.join('%5s' % classes[labels[j]] for j in range(4)))

```



GroundTruth: 7 2 1 0

```

[9]: net = Net()
    net.load_state_dict(torch.load(PATH, weights_only=True))

```

```

[9]: <All keys matched successfully>

```

```
[11]: outputs = net(images)

_, predicted = torch.max(outputs, 1)

print('Predicted: ', ' '.join('%5s' % classes[predicted[j]]
                                for j in range(4)))
```

Predicted: 7 2 1 0

```
[12]: correct = 0
total = 0

with torch.no_grad():
    for data in testloader:
        images, labels = data
        outputs = net(images)
        _, predicted = torch.max(outputs, 1)
        total += labels.size(0)
        correct += (predicted == labels).sum().item()

print(f'Accuracy of the network on the 10000 test images: {100 * correct // total} %')
```

```
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OSError: [Errno 16] Device or resource busy: '.nfs00000000053d2d6800003956'

Accuracy of the network on the 10000 test images: 98 %

```

```

[ ]: correct_pred = {classname: 0 for classname in classes}
total_pred = {classname: 0 for classname in classes}

with torch.no_grad():
    for data in testloader:
        images, labels = data
        outputs = net(images)
        _, predictions = torch.max(outputs, 1)
        for label, prediction in zip(labels, predictions):
            if label == prediction:
                correct_pred[classes[label]] += 1
                total_pred[classes[label]] += 1

for classname, correct_count in correct_pred.items():
    accuracy = 100 * float(correct_count) / total_pred[classname]
    print(f'Accuracy for class: {classname:5s} is {accuracy:.1f} %')

```